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2D and 3D anatomical analyses of hand dimensions for custom-made gloves

A. Yu [a], K.L. Yick [a,*], S.P. Ng [b], J. Yip [a]

[a] Institute of Textiles and Clothing, The Hong Kong Polytechnic University, Hung Hom, Hong Kong
[b] Hong Kong Community College, The Hong Kong Polytechnic University, Hong Kong

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ABSTRACT

Measuring hand anthropometric data for the development of good-fitting gloves is crucial. In pursuing higher accuracy in hand anthropometric measurements, scanning of hand surfaces with the aids of image analysis system to acquire measurements is an alternative to the manual methods. This study proposes a new hand measuring approach by using 2D and 3D scanning which are evaluated through comparisons of manual measurements. Thirty-three dimensions are measured by using (1) tape and calliper measurement; (2) 2D image analysis; (3) 3D image analysis based on ten captures; and (4) 3D image analysis based on three captures, respectively. Repeated-measures ANOVA, correlation analysis and RMSE are used to examine the results. The hand dimensions obtained from the four methods are highly linearly correlated. Hand data taken from 3D image analysis has no significant difference compared with manual measurements on hand and wrist circumferences, length and breadth dimension, regardless of the number of captures.

1. Introduction

Gloves are commonly used to protect workers from various hazards in the work environment. However, people often prefer to work with bare hands due to the ill-fitting and obsolete functional design of gloves. Custom-made gloves were mainly found in the area of pressure therapy and required by experts in certain sports. Pressure therapy gloves are engineered to apply adequate pressure to hand and fingers with the aims to increase the rate of scar maturation, prevent contracture formation, and enhance cosmetic appearance without impairing circulation (Leung et al., 2010; Fricke et al., 1999; Rappoport et al., 2008; Silfen et al., 2001). The gloves must fit like a second skin to patients. To optimise the effectiveness and practical use of the gloves, an accurate and efficient measurement of hand anthropometric dimensions for pattern development is crucial.

In making custom-made gloves, direct measurement tools such as flexible measuring tape, callipers, measuring boards and rulers are traditionally used to obtain hand dimensions for glove design and pattern development (Garrett, 1971). They are generally small in size, easy to use, inexpensive, but result in low accuracy due to the complex anatomy and curvature of the wrists, thumbs, fingers and related joints. As the human body is supple and flexible, the tension of a measuring tape and the curvatures of corresponding landmarks and positions can influence the hand dimension results, which lead to poor repeatability and large variances, particularly when measurements are taken by different people. Currently, the inaccurate hand dimensions due to the lack of suitable measuring equipment in hospitals result in re-measurements and repeated adjustments for glove fitting, thus adversely affecting the efficacy of the pressure therapy treatment. The current glove making process is therefore very time-consuming and alternatives have been involving lots of "trial-and-error", based on the experience of individual practitioners.

More recently, apart from direct measurement methods, indirect methods such as three-dimensional (3D) image analyses have been widely adopted for taking body dimensions in the design of various fashion and medical products (Yu and Tu, 2009; Yu et al., 2003; Lu et al., 2010; Hsu and Yu, 2010; Lu and Wang, 2008). 3D body scanning captures several images of a human body from various angles which are then combined to form a 3D image. Body dimensions could be obtained from this 3D image with the aid of a computer program. However, certain techniques are required in the image capturing process, which are landmarking, 3D image registration and hand dimension analysis (Kouchi and Mochimaru, 2011). This directly affects the accuracy of the results. As most of the 3D scanning technologies with computer program for image analysis are specifically designed for measuring the entire body dimensions, the required hand dimensions for making custom-made gloves such as finger length and circumference could not

* Corresponding author. Tel.: +852 2766 6551; fax: +852 2773 1432.
E-mail address: tcyick@inet.polyu.edu.hk (K.L. Yick).

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be obtained from the 3D body scanners available in market. The costly equipment and maintenance of 3D body scanner is another shortcoming.

On the other hand, multi-camera photogrammetric systems based on two-dimensional (2D) images for body measuring have been developed in various studies and used in different medical applications such as monitoring facial shape, skin wound, teeth abrasion, etc. (Lin and Wang, 2011; Meunier and Yin, 2000; Deng et al., 2007; Meintjes et al., 2002). Based on the measuring principle of photogrammetry, the surface data of hand can be acquired by registering 2D hand images from different viewing angles within a second. The digital cameras are relatively less expensive. However, the images obtained can be easily affected by the number of images being registered, viewing angle and the lighting (Lau and Armstrong, 2011).

The objective of the present study is to propose a new approach of taking hand anthropometric measurements by using 2D and/or 3D image analyses and compare against the traditional direct measurement method. This research work aims to provide a useful reference for the development of hand anthropometry, thus adding a new dimension to medical clothing and/or high quality custom-made gloves that perfectly fit the hands of sport players and professionals in order to achieve better performance and protection.

2. Method

2.1. Participants

A total of 10 healthy volunteers, 5 males and 5 females, were invited to participate in this study. Their ages were between 20 and 28 (mean: 22.5, SD: 2.77). The ranges of the hand circumference, hand length and digit 3 finger root circumference of the left hands within the male group and the female group of participants were stated below.

- Hand circumference: male (mean: 204 mm, SD: 1 mm); female (mean: 186 mm, SD: 8 mm)
- Hand length: male (mean: 176 mm, SD: 5 mm); female (mean: 170 mm, SD: 10 mm)
- Digit 3 finger root circumference: male (mean: 64 mm, SD: 7 mm); female (mean: 60 mm, SD: 5 mm)

Their hand sizes were within a range that the variations between different hand measuring methods can be readily compared.

2.2. Methodology of measurement

Instead of scanning the real hands of the volunteers, plaster hand models were used in this study. A plaster hand model was first made for each left hand of the 10 volunteers. Participants were asked to widely open their hands and let their fingers spread far apart from one another to take a negative impression. Plaster was then added to create a hand model. Dimensions of real hands and plaster hand models were measured respectively by using measuring tape and calliper. Results are presented in Table 1. Owing to the posture changes, muscular movements and tissue compression on real hands, dimension variations between the real hands and plaster hands were found. To minimise the potential discrepancies on real hands during the measuring and/or image scanning processes, plaster hand models are therefore used in this study for proper evaluation of different hand anthropometric measuring methods.

Table 1
Hand sizes of the participants.

Hand dimension	Measurements (mm)			
	Real hand		Plaster hand	
	Mean	S.D.	Mean	S.D.
Digit 3 distal interphalangeal joint circumference	46.50	3.65	50.00	4.14
Digit 3 finger root circumference	61.25	5.52	67.90	6.30
Hand circumference (pass over metacarpale of Digits 2 and 5)	191.70	10.90	195.80	11.98
Circumference between thumb and palm (around the crease mark)	98.05	10.24	100.30	9.48
Wrist circumference	157.20	12.13	167.50	12.66
Digit 3 Finger length (from fingertip to root)	73.52	2.28	73.23	2.39
Digit 3 tip to wrist-crease length/Hand length	171.50	9.46	171.65	9.36
Length of root of index-finger to the root of thumb	35.46	5.74	34.42	3.89
Length of root of little finger to outer wrist	66.41	6.90	65.17	5.39
Length of root of thumb to the inner wrist	56.31	5.40	56.22	5.11

With reference to the hand dimension requirement for the pattern of a pressure therapy glove and a study by Kwon et al. (2009) which was based on a 1988 US Army hand anthropometric survey report (Greiner, 1991) to determine the key dimensions for a glove sizing system, the present study defines 49 landmarks (Fig. 1) and 33 dimensions (Table 2) for examination. Each of the plaster hands were measured by a traditional direct anthropometric measurement method (DM) and three indirect measurement methods, namely, Methods IM-I, IM-II and IM-III respectively. DM was carried out by using a measuring tape (with a minimum scale of 1 mm and thickness of 0.4 mm) to take the circumference dimensions and a digital calliper (with an accuracy of 0.01 mm) to measure the length and breadth dimensions. Measurements were taken on the palmar surface of the hand following ISO 7250-1:2008 standard. It is noted that the fingers were spread out during the scanning process so as to build a complete 3D hand image for measuring the required hand dimensions such as circumferences of fingers and related joints. As the plaster hand models were rigid and fixed in shape, the influence of hand posture on the results of hand measurement could be neglected.

2.2.1. 2D image analysis (Method IM-I)

In order to minimise the time for calibration and set-up of a camera for 2D image capturing, a colour scanner is used in this work. An Epson Perfection 1240U domestic colour scanner was adopted to scan the plaster hands to generate a 2D image for each hand. A plaster hand was put directly on the scanning screen with the palm facing down and the image was taken as illustrated in Fig. 2. The scanning process could be completed in 2 min. The scanned images were saved as bitmap files with a resolution of 300 dpi and then imported to a computer software (CorelDRAW X5, Corel Corporation) to measure the required hand dimensions. The scanner was calibrated beforehand to ensure the measurements obtained from the 2D image were identical to the scanned object. A standard grid paper with the grid size of 1×1 cm^2 was first scanned. The image was then imported to the software and the size of the grid was measured with a minimum scale of 0.1 mm. The amount of variations obtained from the grid images was less than 1% which was within the tolerance of measurement. As the cover of the scanner could not be completely closed during the hand scanning, the scanning process was conducted in a dark environment to avoid the potential influence of light intensity.

It is noteworthy that the circumferences of the thumb, fingers, and related joints could not be taken based on 2D images. A calliper

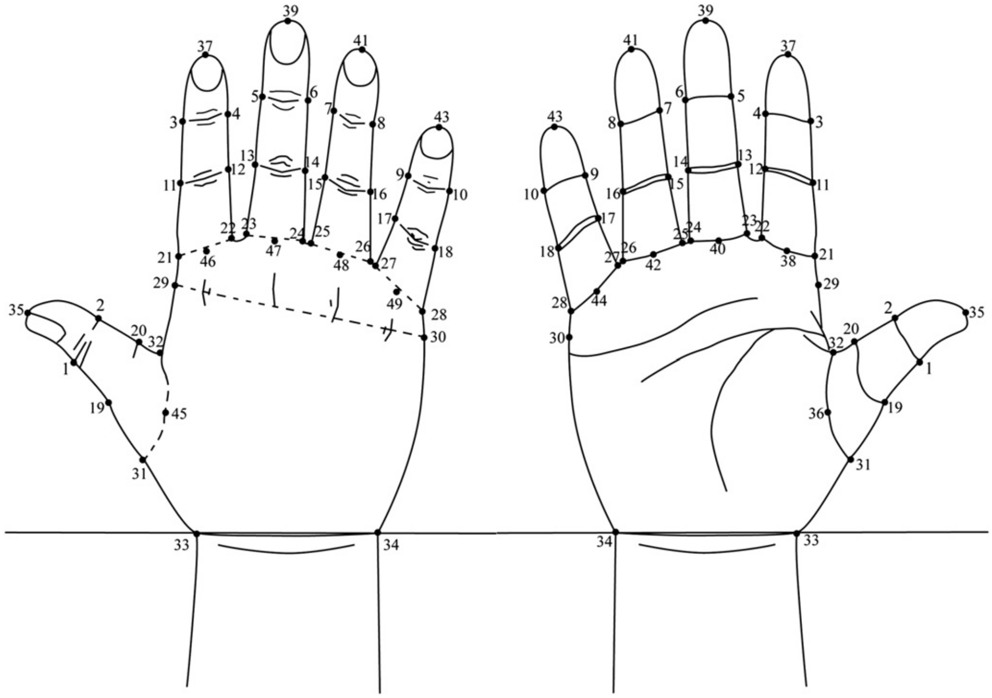

Fig. 1. Landmarks for hand measurement.

Table 2
Hand dimensions measured in the present study.

Type	No.	Dimension	Landmarks
Circumference	C1	Digit 1 interphalangeal joint circumference	Across 1 and 2
	C2	Digit 2 distal interphalangeal joint circumference	Across 3 and 4
	C3	Digit 3 distal interphalangeal joint circumference	Across 5 and 6
	C4	Digit 4 distal interphalangeal joint circumference	Across 7 and 8
	C5	Digit 5 distal interphalangeal joint circumference	Across 9 and 10
	C6	Digit 2 proximal interphalangeal joint circumference	Across 11 and 12
	C7	Digit 3 proximal interphalangeal joint circumference	Across 13 and 14
	C8	Digit 4 proximal interphalangeal joint circumference	Across 15 and 16
	C9	Digit 5 proximal interphalangeal joint circumference	Across 17 and 18
	C10	Digit 1 finger root circumference	Across 19 and 20
	C11	Digit 2 finger root circumference	Across 21 and 22
	C12	Digit 3 finger root circumference	Across 23 and 24
	C13	Digit 4 finger root circumference	Across 25 and 26
	C14	Digit 5 finger root circumference	Across 27 and 28
	C15	Hand circumference (pass over metacarpale of Digit 2 and 5)	Across 29 and 30
	C16	Circumference between thumb and palm	Across 31 and 32
	C17	Wrist circumference	Across 33 and 34
Length	L1	Digit 1 Finger length (from fingertip to root)	From 35 to 36/45
	L2	Digit 2 Finger length (from fingertip to root)	From 37 to 38/46
	L3	Digit 3 Finger length (from fingertip to root)	From 39 to 40/47
	L4	Digit 4 Finger length (from fingertip to root)	From 41 to 42/48
	L5	Digit 5 Finger length (from fingertip to root)	From 43 to 44/49
	L6	Digit 1 tip to wrist-crease length	From 35 pass through 36/45 to the line across 33 and 34
	L7	Digit 2 tip to wrist-crease length	From 37 pass through 38/46 to the line across 33 and 34
	L8	Digit 3 tip to wrist-crease length	From 39 pass through 40/47 to the line across 33 and 34
	L9	Digit 4 tip to wrist-crease length	From 41 pass through 42/48 to the line across 33 and 34
	L10	Digit 5 tip to wrist-crease length	From 43 pass through 44/49 to the line across 33 and 34
	L11	Length of root of index-finger to the root of thumb	From 21 to 20
	L12	Length of root of little finger to outer wrist	From 28 to 34
	L13	Length of root of thumb to the inner wrist	From 19 to 33
	L14	Palm length (perpendicular distance from the base of digit 3 to the wrist crease base line)	From 40 to the line across 33 and 34
Breadth	B1	Hand breadth (distance from hand's radial edge to ulnar edge)	From 29 to 30
	B2	Wrist breadth (between the ulnar projection and the radial projection of the distal wrist crease)	From 33 to 34

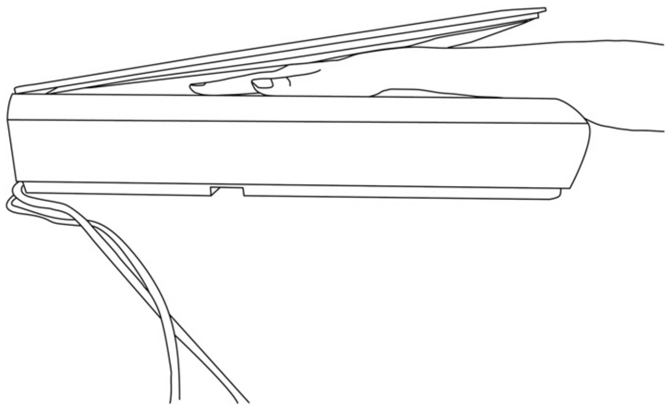

Fig. 2. 2D image taken by a colour scanner.

was therefore used to measure the finger, hand and wrist thicknesses (Fig. 3) along the landmarks. With the width and thickness, the circumference dimensions of the finger, hand and wrist could be predicted by a mathematical model. The circumferences are shaped like an ellipse. Yao et al. (2008) used a super-ellipse curve to simulate human body cross-sections which were regarded as ellipse like. A super-ellipse could better mimic the finger, hand and wrist circumferences than a true ellipse. However, the fitting of data to the super-ellipse and the calculation of its perimeter are complicated (Zhang and Rosin, 2003). Therefore, in this study, Ramanujan II approximation of the ellipse perimeter is chosen to calculate the circumferences.

$$C \approx \pi(a+b)\left(1+\frac{3\left(\frac{a-b}{a+b}\right)^2}{10+\sqrt{4-3\left(\frac{a-b}{a+b}\right)^2}}\right)$$

where C is the circumference of the ellipse, and a is the semi-major and b is the semi-minor axes of the ellipse.

2.2.2. 3D image analysis

To scan the 3D images of the plaster hands, a portable 3D laser scanner (NextEngine Inc.) was used and the precise hand measurements can be obtained during clinical practices in various locations in a hospital. The scanner was operated with a scanning software, ScanStudio HD (NextEngine Inc.). After a 3D image of the hand was obtained, the hand dimensions could be extracted by Reserve engineering software, RapidForm 2006 (INUS Technology Inc.). Two ways of 3D image capture were investigated in this study.

2.2.2.1. 3D image combined from ten captures (Method IM-II). To form a complete 3D image of a plaster hand from the 3D scanner, several shots that could capture all the parts of the plaster hand were essential. In Method IM-II, a plaster hand was put onto a rotatable disc that was connected to a 3D scanner with the fingers pointing to the sky (Fig. 4). The rotatable disc was placed 17 inches away from the 3D scanner to ensure that the images were in focus. The images were taken in the quality of standard configuration which took about 2 min to complete the capturing of each image.

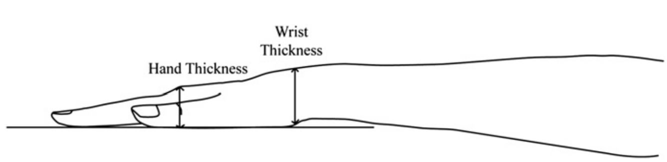

Fig. 3. Hand and wrist thickness measurement.

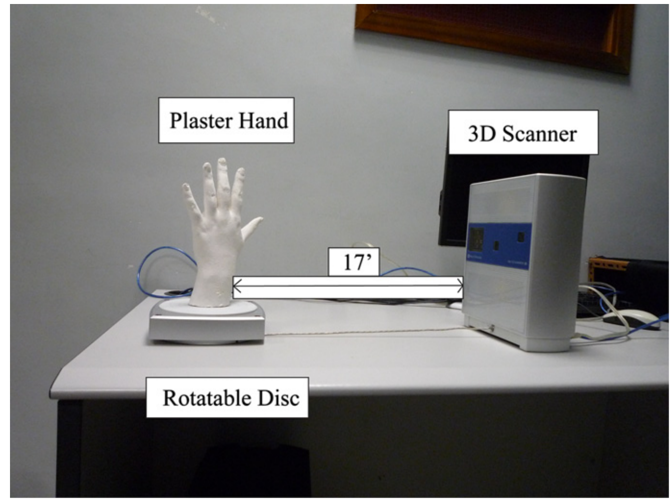

Fig. 4. Setting for 3D scanning of a plaster hand by Method IM-II.

The 3D scanner was simultaneously operated with the disc and automatically took images after each rotation. The disc rotated 36° after each image capture and stopped once returned to the original position. Therefore, ten captures from different angles of view can be obtained and combined into a 3D image with completed surface data by the scanning software.

2.2.2.2. 3D image combined from three captures (Method IM-III). Speed and user-friendliness are crucial elements for the scanning method to be acceptable. A simplified approach of 3D image analysis by combining three captures from three viewing angles was therefore proposed in this study. Method IM-III captured the 3D surface data of a plaster hand on a flat surface with the palm facing down. The 3D scanner only shot the back side of plaster hand three times from three different angles to obtain a top, a left side and a right side view (Fig. 5) with the support of a tripod. The distance between the 3D scanner and the hand was kept at a constant length of 17 inches. The 3D surface data obtained from the registration of the three captures was not complete and could only show the back of the hand as images on the palm side were missing. This method aims to minimise the number of hand images and scanning time to get a 3D hand image of dorsal surface. It also allows participants to rest their hands comfortably, thus minimizing the undesired hand movements. However, hand dimensions can still be measured and estimated with this 3D surface data. There were measurement landmarks on both the palm and back of the hand. When obtaining the length measurement of a hand, the basis of Method IM-III was

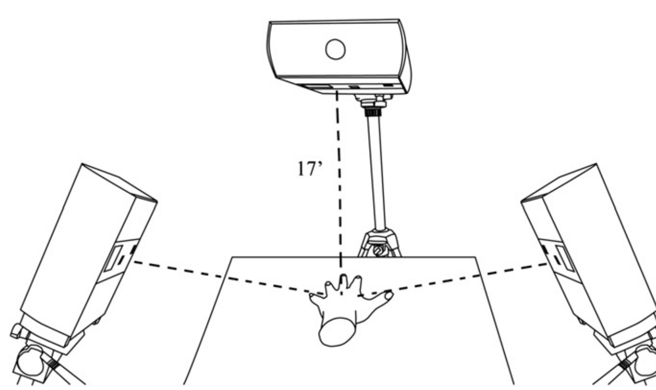

Fig. 5. Setup for 3 image capturing with 3D scanner by Method IM-III.

on the landmarks on the back of the hand while the other three methods worked according to the landmarks on the palm. This is because the image captured by Method IM-III could only show the back of the hand. To measure the circumference of the fingers, it was assumed that fingers are cylindrical and based on the arc length and shape to determine their circumferences. The palm and wrist circumferences were calculated by doubling the arc length of the palm measured from the back of the hand.

2.3. Statistical analysis

In this study, the hand dimension results obtained from the four methods were analysed by using IBM SPSS Statistic software. Repeated-measures analysis of variance (ANOVA) was adopted to examine the hand dimensions obtained from the four methods for the analysis of variance. A correlation analysis was also used to determine the relationship between the different measuring methods. The root mean square error between DM and the image analysis methods were calculated to present the discrepancies.

3. Results

3.1. Images captured

As shown in Fig. 6, a 2D image of a plaster model is obtained. The corresponding length and breadth dimensions are easily extracted. Figs. 7 and 8 show 3D images obtained by Methods IM-II and IM-III. The results reveal that Method IM-II with 10 captures can successfully establish a complete and smooth 3D hand model, whilst the 3D hand model that combines 3 captures by Method IM-III is relatively incomplete. The image quality is relatively poor and blurry, particularly along the hand edges.

3.2. Repeated-measures ANOVA on hand dimensions obtained by different measuring methods

Repeated-measures ANOVA can examine the relationship between a repeated-measures categorical variable and a quantitative dependent variable (Salkind and Green, 2011). With the use of repeated-measures ANOVA, the significance of the differences among the different means of hand measurement from the four methods could be tested. Descriptive statistics, measures of skewness and kurtosis, histogram and normal Q–Q plots were examined to check if the assumption of normality is fulfilled. Dimensions C10, C12, L1, L5, L7 and L9 were skewed, data transformation was conducted. Huynh–Feldt corrections were applied in this study for the cases that Mauchly's test of sphericity assumption was violated. The results of the repeated-measures ANOVA are listed in Table 3. They indicate that the hand measuring methods have significant impacts on the measured hand dimensions. In this study, 26 out of the 33 hand dimensions have significant mean differences ($p < 0.05$) amongst the four methods. Only (1) hand circumference, (2) circumference between thumb and palm, (3) wrist circumference, (4) digit 1 finger length, (5) digit 5 finger length, (6) digit 2 tip to wrist-crease length and (7) digit 3 tip to wrist-crease length have no significant differences.

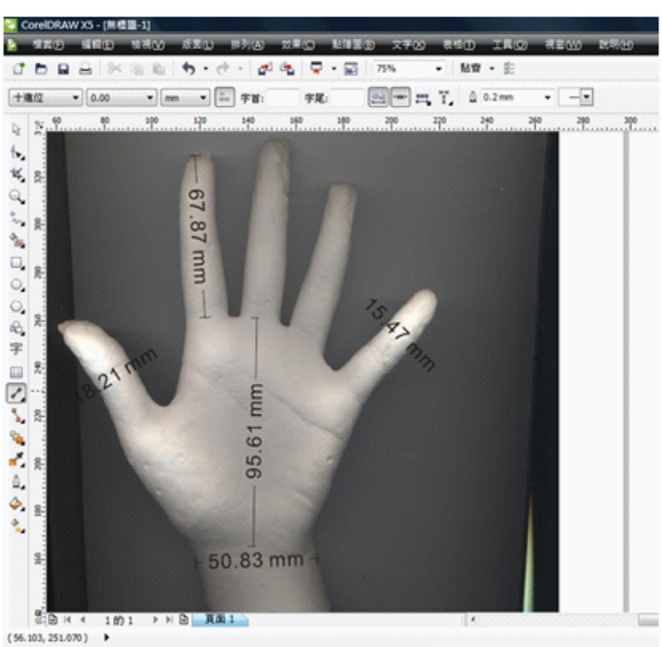

Fig. 6. Hand dimension measuring on a 2D image by using CorelDRAW X5.

Sidak pairwise comparison tests were chosen as the post hoc test to further examine the relationship between each pair of means of the measurements from these methods. Table 4 shows the pairwise comparisons of significant differences on hand dimensions measured by the four methods. At least half of the hand dimensions in each comparison have no significant differences ($p > 0.05$). Most of the length dimensions of the hands do not have significant differences between any pair in the comparisons. However, significant differences between DM and Method IM-I and between DM and Method IM-II are observed in the measurement of the finger circumferences. There are 27 hand dimensions measured by Method IM-III which showed no significant differences when compared with those by DM. However, only 18 and 17 hand dimensions show no significant differences between DM and Method IM-I or IM-II respectively. Yet, there are 24 hand measurements which show no significant differences between Methods IM-II and IM-III.

When examined by repeated-measures ANOVA, the type of hand measuring method has significant differences on the measurements of hand circumference ($F(3405.389) = 103.791$, $p = 0.000$, $\eta^2 = 0.38$), length ($F(3405.977) = 13.233$, $p = 0.000$, $\eta^2 = 0.087$) and breadth ($F(3, 57) = 37.164$, $p = 0.000$, $\eta^2 = 0.662$). However, pairwise comparisons of the significant differences in hand circumference, length and breadth (Table 5) do not appear to be significantly different in length and breadth for DM, and Methods IM-II and IM-III. On the other hand, Method IM-I presents significant differences in the circumference, length and breadth when compared with DM, Method IM-II or IM-III.

3.3. Correlation analysis

Upon examination, the Pearson correlation coefficients could reflect the strength of the linear relationship between each pair of measuring method. The Pearson correlation coefficients between the four methods of measurement are presented in Table 6. It can be seen that the correlation coefficients for the relationships between the four principal study variables range from 0.993 to 0.997 ($n = 330$, $p < 0.001$ (two-tailed)) which indicate that they are highly positively correlated with one another. Method IM-II has a slightly higher correlation with DM than Methods IM-I and III. The hand dimension measured by an analysis of the 3D image from the combination of ten captures is also highly correlated with that from three captures, $r = 0.997$, $n = 330$, $p < 0.001$ (two-tailed). The bivariate scatterplots in Fig. 9 indicate that Methods IM-II and IM-III are highly linearly correlated. The results also revealed that Methods IM-I, IM-II and IM-III have a close linear relationship with DM.

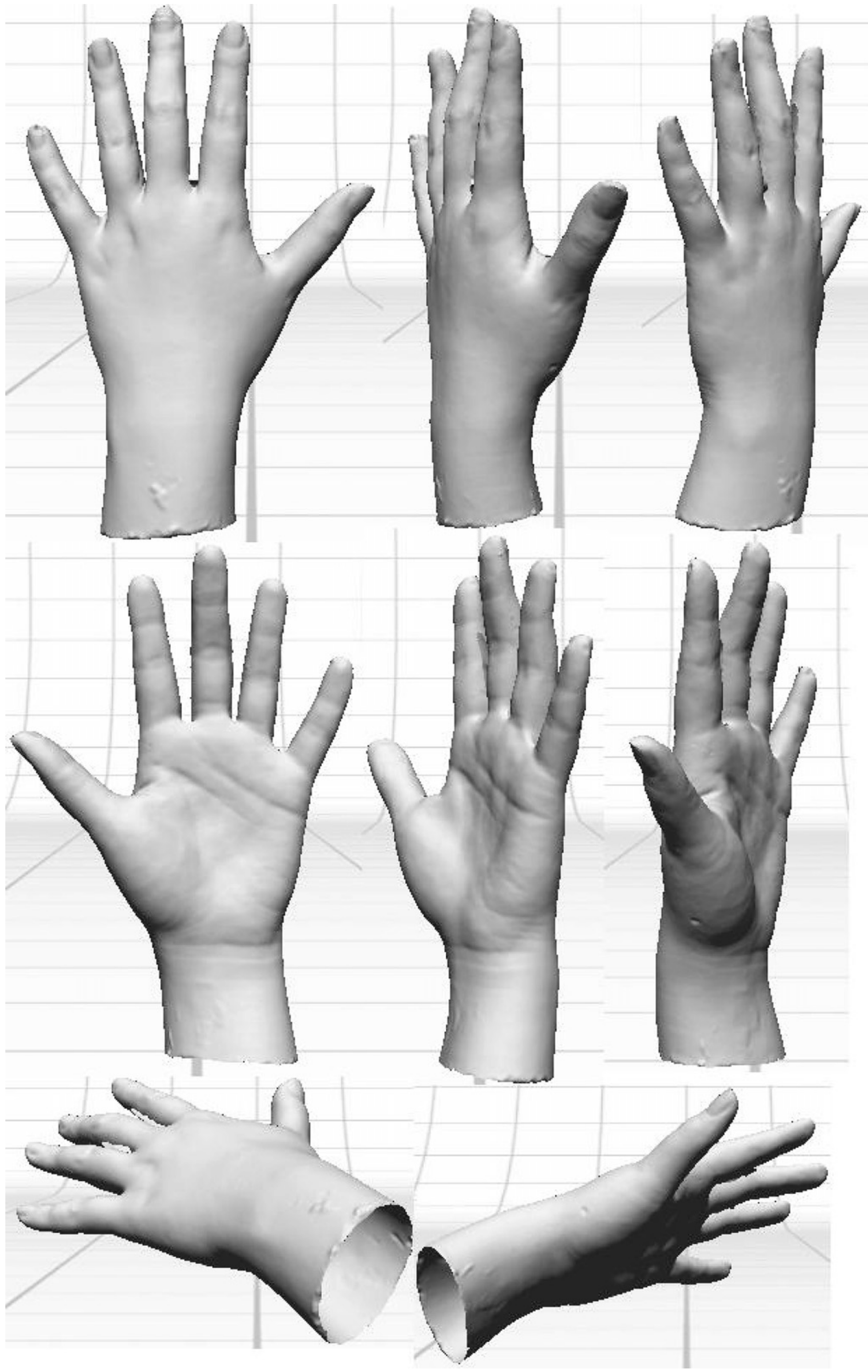

Fig. 7. Various views of a 3D image taken by Method IM-II.

3.4. Root mean square error

In order to understand the hand dimension discrepancies of the image analysis methods, the root mean square error (RMSE) between DM and each of image analysis method was calculated respectively. As shown in Table 7, small differences are recorded in hand dimensions between DM and each of the image analysis method. As compared with the dimensions obtained by DM, Method IM-I has the smallest differences at finger length measurements, the RMSE range from 0.85 mm to 3.05 mm. In the case of Method IM-III, the corresponding finger length differences are relatively high, the RMSE range from 2.23 mm to 4.38 mm.

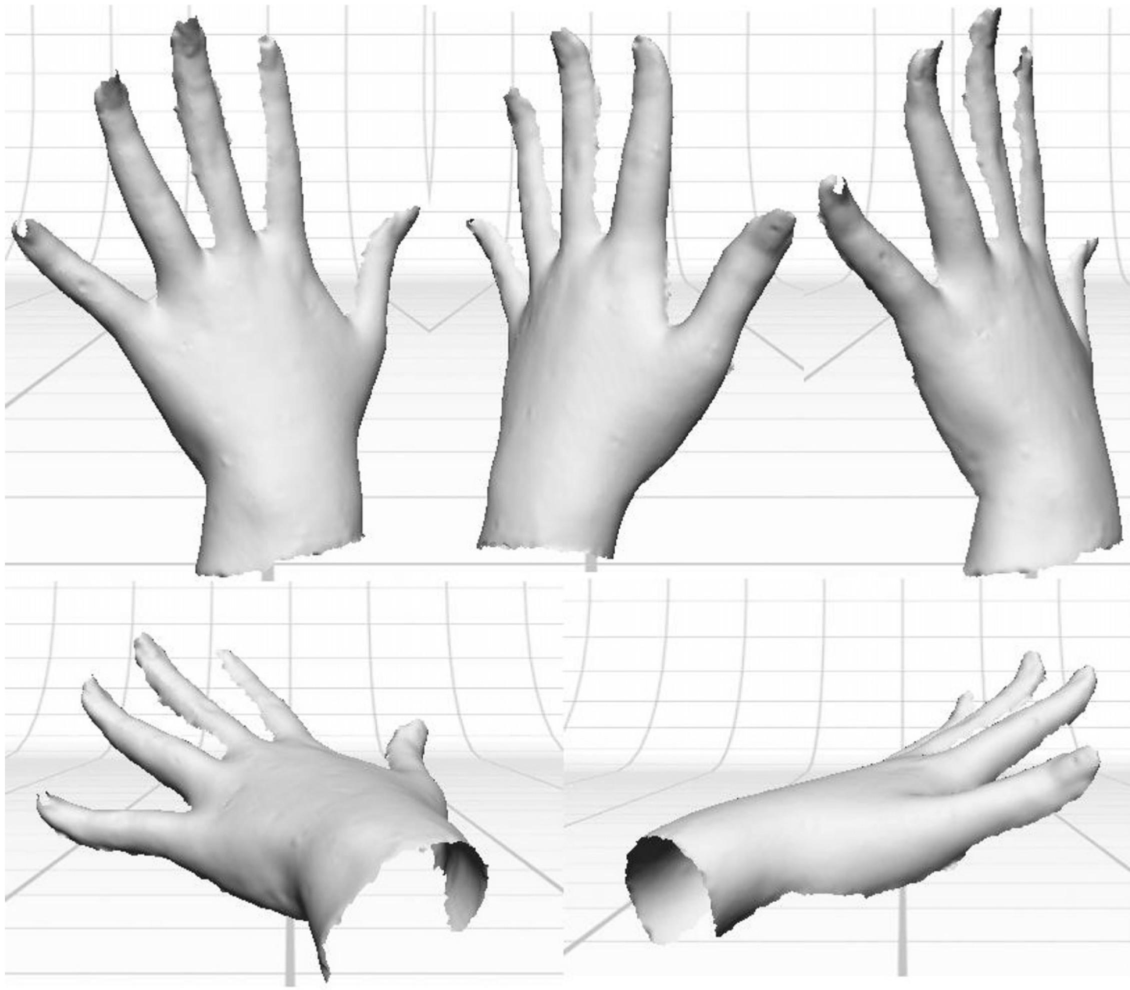

Fig. 8. Various views of a 3D image taken by Method IM-III.

However, the dimension differences between Method IM-III and DM obtained at the finger interphalangeal joint circumferences, hand breadth and wrist breadth are relatively small, the RMSE range from 0.82 mm to 3.52 mm. The corresponding differences at hand and wrist breadths obtained from Method IM-I are relatively high, which RMSE range from 1.61 mm to 7.57 mm. Amongst the 3 image analysis methods, Method IM-II exhibits the least discrepancies at the finger root circumferences, hand and wrist circumferences, fingertip to wrist lengths and palm length as compared with DM.

4. Discussion

The result of the repeated-measures ANOVA indicated that there are significant differences among the four methods of hand measurement on most of the hand dimension measuring, even though the pairwise comparison showed that there is no significant difference in more than half of the hand measurements between any two methods. The differences shown in repeated-measures ANOVA might be explained by the geometry of the laser system which can result in errors in the order of millimetres (Witana et al., 2006). Other than that, the 2D and 3D images obtained for the use of analysis provided different degrees of information to obtain hand measurements. The 3D image from Method IM-II delivered the most complete image and largest amount of hand information. As the images could not show all parts of the hand in 2D and 3D imaging by Method IM-I and IM-III, estimation and calculation were needed to compensate for the incomplete areas, especially in obtaining circumference dimensions. With less hand information obtained from the image, more predictions were needed and hence, a greater chance of leading to variance in the results. To enhance the accuracy of the hand anthropometric data, further studies on soft hand models or real hands will be needed to optimise the prediction model used in various image analysis methods. A further investigation of taking hand anthropometric measurements by using 2D scanning based on real hands has been done on 80 subjects. Results are presented in the next paper.

Method IM-I experienced significant differences with the three other types of measuring methods on hand circumference, length, and breadth dimension measurements. It had a relatively large RMSE with DM when compared with Method IM-II and IM-III. Nevertheless, it still performed certain hand measurements with no significant differences when compared with other methods as shown in Table 4. As a plaster hand was rigid and its posture was fixed, the palm and fingers could not completely come into contact with the scanning screen of the scanner which created a small amount of distance from the screen at certain points. Any distance between the plaster hand and the scanning screen could affect the hand size which appeared in the 2D image taken. The points of the hand which had some distance from the screen were scattered, and

Table 3
Summary of a variance analysis of the significant differences on hand dimensions as measured by the four methods.

Dimension	*F*	*df*	*p(sig.)**	η2
Digit 1 interphalangeal joint circumference	12.87	3	0	0.59
Digit 2 distal interphalangeal joint circumference	13.92	3	0	0.61
Digit 3 distal interphalangeal joint circumference	12.22	1.85	0	0.58
Digit 4 distal interphalangeal joint circumference	6.73	3	0	0.43
Digit 5 distal interphalangeal joint circumference	9.71	3	0	0.52
Digit 2 proximal interphalangeal joint circumference	18.08	3	0	0.67
Digit 3 proximal interphalangeal joint circumference	24.09	3	0	0.73
Digit 4 proximal interphalangeal joint circumference	18.7	3	0	0.68
Digit 5 proximal interphalangeal joint circumference	7.59	2.29	0	0.46
Digit 1 finger root circumference	9.68	2.32	0	0.52
Digit 2 finger root circumference	34.67	3	0	0.79
Digit 3 finger root circumference	27.85	1.94	0	0.76
Digit 4 finger root circumference	26.1	1.66	0	0.74
Digit 5 finger root circumference	12.81	1.72	0	0.59
Hand circumference (pass over metacarpale of Digits 2 and 5)	0.6	2.08	0.57	0.06
Circumference between thumb and palm (around the crease mark)	1.93	3	0.15	0.18
Wrist circumference	1.38	3	0.27	0.13
Digit 1 Finger length (from fingertip to root)	2.87	3	0.06	0.24
Digit 2 Finger length (from fingertip to root)	3.93	3	0.02	0.3
Digit 3 Finger length (from fingertip to root)	7.03	3	0	0.44
Digit 4 Finger length (from fingertip to root)	7.02	1.8	0.01	0.44
Digit 5 Finger length (from fingertip to root)	1.74	1.65	0.21	0.16
Digit 1 tip to wrist-crease length	5.89	3	0	0.4
Digit 2 tip to wrist-crease length	2.81	3	0.06	0.24
Digit 3 tip to wrist-crease length	3.51	1.73	0.06	0.28
Digit 4 tip to wrist-crease length	4.42	3	0.01	0.33
Digit 5 tip to wrist-crease length	6.76	3	0	0.43
Length of root of index-finger to the root of thumb	3.38	3	0.03	0.27
Length of root of little finger to outer wrist	4.5	3	0.01	0.33
Length of root of thumb to the inner wrist	4.43	3	0.01	0.33
Palm length (perpendicular distance from the base of digit 3 to the wrist crease base line)	8.59	2.67	0	0.49
Hand breadth (distance from hand's radial edge to ulnar edge)	15.88	3	0	0.64
Wrist breadth (between the ulnar projection and the radial projection of the distal wrist crease)	26.76	3	0	0.75

* The difference which is not significant at the 0.05 level is highlighted.

the distance was also small, which made it difficult to take into account the dimension extraction. This problem should be reduced with real hands as they are more flexible and can lie on the scanning screen better than a plaster hand. When a 2D palm view image of the real hand is obtained, the length dimensions are believed to be measured more accurately with higher repeatability than DM during which the hand of the subject might not open straight and exhibit muscular movement and posture changes. Further studies are needed on real hands to verify the accuracy of the 2D image analysis method. This method could be put into practice after the measurement model was further optimised. The time for image taking with Method IM-I is much faster than the whole hand dimensions measured by DM. The case is more remarkable when more measurements need to be taken. On-line purchases of custom-made gloves require customers to take hand measurements by themselves with a measuring tape. Any inaccurate measuring can affect the fit of the glove. With the use of 2D analysis, customers could buy tailor-made gloves on-line simply by scanning their hand with a domestic 2D scanner to obtain their hand measurements. This not only can eliminate the possible errors in tape measuring, but also save the time of both customers and manufacturers and simplify the entire selling process.

Both types of 3D image analysis exhibited no significant differences in hand and wrist circumferences and length and breadth dimension measurements with the direct measuring method. These demonstrate that both types of 3D analysis could perform well in measuring such dimensions. The fit of a custom-made glove mainly depends on the hand measurements applied in pattern making. Accurate and precise hand measurements cannot be omitted in the designing and developing process. Gloves are also extensively used in industrial work for protection or improving the friction between hand and material surfaces. A study by Tsaousidis and Freivalds (1998) pointed out that when wearing a glove, the grip force of the hand would be reduced in both its maximum value and the rate of its development. This is due to some of the work done by hand is stored as elastic energy in the creases of the glove when the compression forces increase. In other words, the reduction of the grip force caused by a glove can be diminished if the glove perfectly fits the wearer hand so as to cut down the amount of creases that appear on the glove. Yick et al. (2011) proved that the use of 3D scanning to capture images for evaluation can have high precision and reproducibility which are lacking in DM. 3D image analysis could fulfil the increase in the demands and requirements of custom-made gloves. Although 3D image analysis methods take more time in image capturing than 2D image analysis methods, much simpler estimation and calculations are involved in extracting the hand dimensions. 3D image analysis can obtain results that are closer to DM with smaller RMSE than 2D image analysis. Once a 3D image was captured, any dimension on the hand at any point as desired could be extracted anytime and anywhere. A complete record of the hand could be kept in this way to keep track of the changes over a period of time which is especially useful in the

Table 4
Summary of pairwise comparisons of the significant differences in hand dimensions measured by the different measuring methods.

Dimension	p(sig.)*					
	DM VS IM-I	DM VS IM-II	DM VS IM-III	IM-I VS IM-II	IM-I VS IM-III	IM-II VS IM-III
Digit 1 interphalangeal joint circumference	0.01	0	0.13	0.83	0.01	0.26
Digit 2 distal interphalangeal joint circumference	0.01	0	0.28	1	0.14	0.01
Digit 3 distal interphalangeal joint circumference	0.01	0	1	0.42	0.17	0.01
Digit 4 distal interphalangeal joint circumference	0.01	0	0.51	0.9	0.89	0.28
Digit 5 distal interphalangeal joint circumference	0.81	0	1	0.09	0.69	0.01
Digit 2 proximal interphalangeal joint circumference	0.01	0.01	1	0.8	0	0
Digit 3 proximal interphalangeal joint circumference	0	0	0.18	0.01	0	0.05
Digit 4 proximal interphalangeal joint circumference	0	0	0.18	1	0	0.02
Digit 5 proximal interphalangeal joint circumference	0.06	0.06	0.2	1	0.09	0.51
Digit 1 finger root circumference	0.07	0.67	0.01	0.02	1	0.01
Digit 2 finger root circumference	0	0	0.05	0.01	0	0.7
Digit 3 finger root circumference	0	0	0.01	0.01	0	0.98
Digit 4 finger root circumference	0	0	0	0.1	0.1	0.54
Digit 5 finger root circumference	0.02	0.01	0.02	0.26	0.74	0.92
Hand circumference (pass over metacarpale of Digits 2 and 5)	1	0.46	0.71	1	0.99	0.99
Circumference between thumb and palm (around the crease mark)	0.98	0.03	0.65	0.84	1	0.94
Wrist circumference	0.48	0.49	0.91	0.98	0.94	1
Digit 1 Finger length (from fingertip to root)	0.99	1	0.07	0.99	0.1	0.17
Digit 2 Finger length (from fingertip to root)	1	0.97	0.22	0.98	0.09	0.32
Digit 3 Finger length (from fingertip to root)	0.94	0.54	0.04	1	0.13	0.17
Digit 4 Finger length (from fingertip to root)	1	1	0.05	1	0.03	0.19
Digit 5 Finger length (from fingertip to root)	0.25	0.05	1	0.69	0.98	0.76
Digit 1 tip to wrist-crease length	0.03	0.03	0.72	1	0.55	0.24
Digit 2 tip to wrist-crease length	0.87	0.99	0.52	0.97	1	0.2
Digit 3 tip to wrist-crease length	0.34	0.63	0.1	0.57	0.82	0.77
Digit 4 tip to wrist-crease length	1	0.19	1	0.05	0.78	0.03
Digit 5 tip to wrist-crease length	0.03	0.7	0.95	0.04	0.08	1
Length of root of index-finger to the root of thumb	0.97	0.42	0.96	0.01	0.36	0.8
Length of root of little finger to outer wrist	1	0.07	0.31	0.05	0.36	0.98
Length of root of thumb to the inner wrist	0.81	1	0.11	0.63	0.03	0.28
Palm length (perpendicular distance from the base of digit 3 to the wrist crease base line)	0.4	1	0.03	0.45	0.36	0.01
Hand breadth (distance from hand's radial edge to ulnar edge)	0	0.01	0.06	0.11	0.02	0.94
Wrist breadth (between the ulnar projection and the radial projection of the distal wrist crease)	0	0.99	1	0	0	0.99

* Adjustment for multiple comparisons: Sidak. The difference which is not significant at the 0.05 level is highlighted.

medical area, for example, to understand the healing progress of burn scars after wearing a pressure therapy garment. It is also noteworthy that the approach of 3D image analyses is a non-contact hand measurement method that pains and related complications of burn scars caused by the use of measuring tape can also be minimised. The tissue compression on real hands which occurs in tape measurement leading to poor repeatability can be avoided in 3D image analysis methods.

The application of Method IM-II can obtain a complete image of a hand, but the time for image capturing is lengthy. The ANOVA results and RMSE with DM indicated that the accuracy of Method IM-III is not worse than that of Method IM-II. Method IM-III can be more practical in measuring the dimensions of a real hand as it takes less time for image taking and a hand can lay on a flat surface during image capturing which minimises unwanted movements. Method IM-III could be a choice for hand measuring in the custom-made glove market. This is particular important for patients with hand burns. Local practitioners indicate that the back of hand is a very common site for burns and scald injuries because clenching fists is one of the human reflex actions when people are in danger. The hypertrophic scars are therefore commonly found on the back of hands. The irregular shape of hypertrophic scars should be carefully and accurately measured so as to formulate custom-made therapy glove pattern. A 3D image of the dorsal surface of hand being obtained from Method IM-III is therefore crucial to prescribe adequate amount of pressure on the scar regions. As compared with Method IM-I which can effectively take hand measurements based on the palmar surface of hand, Method IM-III is desirable in medical applications for acquiring irregular hand shape changes and monitoring treatment progress. Method IM-I is suggested for the purpose of making high quality custom-made gloves for optimal fitting and comfort. Additionally, a new approach of combining Method IM-I and IM-III is suggested. The hand surface data on the palm side is obtained by Method IM-I and the corresponding data on the back of hand is captured by Method IM-III. The results of RMSE revealed that the high discrepancies at finger length dimensions between DM and Method IM-III can be considerably reduced by using Method IM-I. This new approach can therefore compensate the missing surface data in Method IM-III but being more user-friendly and less time-consuming than Method IM-II.

The Pearson correlation coefficients and the bivariate scatterplot presented that each hand measuring method is highly linearly correlated with one another, even though there are significant

Table 5
Summary of pairwise comparisons of the significant differences in hand circumference, length and breadth measured by different measuring methods.

	p(sig.)*		
	Circumference	Length	Breadth
DM VS IM-I	0.003	0.000	0.000
DM VS IM-II	0.001	0.997	0.148
DM VS IM-III	0.002	0.994	0.389
IM-I VS IM-II	0.002	0.000	0.000
IM-I VS IM-III	0.002	0.000	0.000
IM-II VS IM-III	0.002	1.000	0.824

* Adjustment for multiple comparisons: Sidak. The difference which is not significant at the 0.05 level is highlighted.

Table 6
Pearson correlation coefficients between hand dimensions measured by different methods.

	DM	IM-I	IM-II	IM-III
DM	1.000	0.993	0.997	0.995
IM-I	0.993	1.000	0.995	0.995
IM-II	0.997	0.995	1.000	0.997
IM-III	0.995	0.995	0.997	1.000

differences among the four methods. This indicates that the differences might be systematic. Therefore, with reference to the study by Witana et al. (2006), a linear regression model could be the way to reduce the discrepancies between the indirect and the direct measuring methods.

In real practice of custom-made glove manufacturing, it is seldom to measure the entire thirty-three hand dimensions but at least the following ten key hand dimensions are required in developing a glove pattern: (1) digit 3 distal interphalangeal joint circumference, (2) digit 3 finger root circumference, (3) hand circumference, (4) wrist circumference, (5) circumference between thumb and palm, (6) digit 3 finger length, (7) digit 3 tip to wrist-crease length/hand length, (8) length of root index-finger to root of thumb, (9) length of root of little finger to outer wrist and (10) length of root of thumb to the inner wrist. The findings reveal that no significant difference exists amongst the four measuring methods on three of key circumference dimensions which are the hand and wrist circumferences and the circumference between thumb and palm. As comparing the five length dimensions for glove making, the results obtained from DM have no significant difference with the 2D and 3D image analyses methods. This implies that, the glove patterns such as palm size and longitudinal dimensions being developed on the basis of the measurements obtained from the 2D and 3D image analyses methods have no significant difference from the traditional direct anthropometric measurement method. However, significant differences in variance were found amongst the four measuring methods in the digit 3 interphalangeal joint and root circumference. Since the finger circumferences obtained from Method IM-I are consistently smaller than the other measuring methods, the gloves being developed on the basis of Method IM-I could be too tight in circumferences of fingers and interphalangeal joints, leading to poor comfort, reduction of gripping force of hand and finger dexterity. Although Method IM-II gives the smallest RMSE with DM in 7 out of 10 measurements, it exhibits the larger RMSE with DM in the remaining 3. Amongst the 2D and 3D image analyses methods, the patterns being developed on the basis of the hand dimensions obtained from Method IM-III are closely

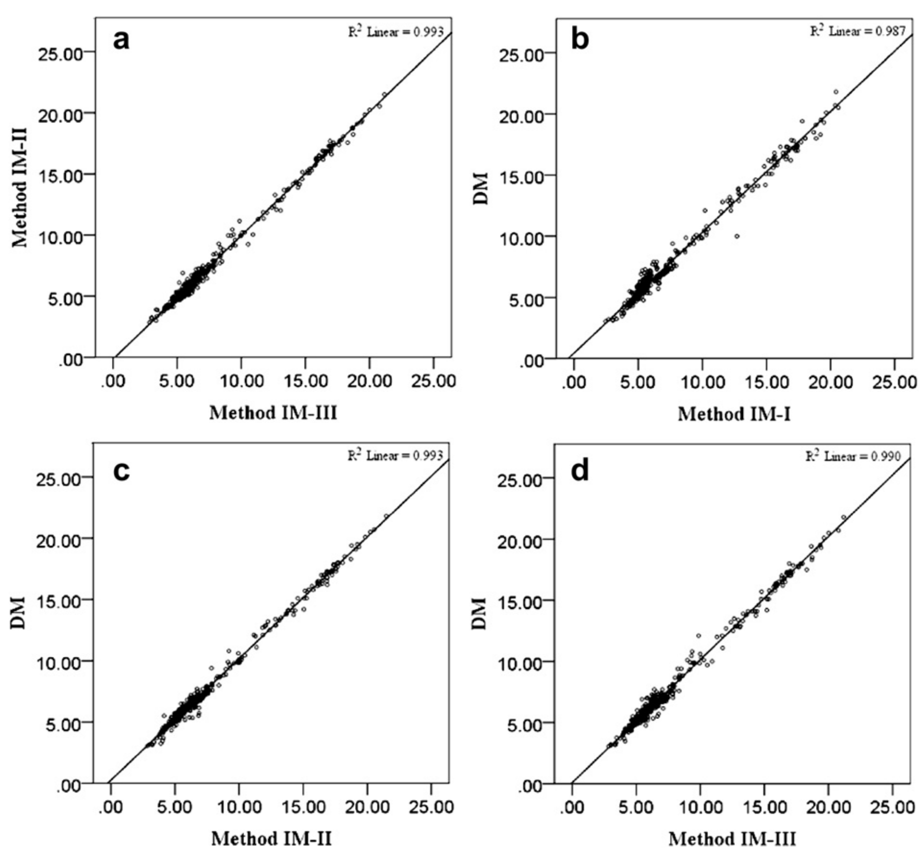

Fig. 9. Bivariate scatterplot that shows the linear relationship between (a) Methods IM-II and IM-III, (b) DM and Method IM-I, (c) DM and Method IM-II and (d) DM and Method IM-III.

Table 7
Hand dimension measurement taken from various measuring methods and the RMSE between DM and each image analysis method.

No.	Measurements (mm)				RMSE (mm) between DM and		
	DM	IM-I	IM-II	IM-III	IM-I	IM-II	IM-III
C1	63.50	59.35	60.40	61.84	5.02	3.50	2.46
C2	50.50	47.05	47.05	49.33	4.12	3.79	1.96
C3	50.00	48.11	47.33	50.12	2.21	2.74	2.16
C4	46.90	44.93	44.28	45.76	2.38	3.03	2.25
C5	43.25	42.63	41.47	43.29	1.61	1.97	0.82
C6	60.75	55.19	56.51	60.76	6.60	4.95	2.76
C7	61.80	54.45	57.51	60.26	7.57	4.64	3.52
C8	58.25	53.44	53.69	56.72	5.60	5.02	2.37
C9	51.80	48.22	48.09	49.63	4.89	5.07	3.43
C10	70.90	65.79	69.14	65.39	6.95	4.18	6.37
C11	68.40	58.08	63.84	65.45	10.81	4.97	3.99
C12	67.90	55.51	63.01	61.64	13.08	5.37	7.48
C13	65.30	57.14	60.49	59.31	9.13	5.05	6.69
C14	59.40	53.70	55.31	54.68	7.22	4.91	5.98
C15	195.80	195.37	194.42	193.75	6.30	2.63	4.77
C16	100.30	97.30	93.66	95.74	12.11	8.51	10.08
C17	167.50	164.13	165.69	166.04	6.53	3.52	4.50
L1	53.04	52.43	53.17	50.77	3.05	3.30	3.13
L2	67.24	67.50	67.91	69.53	1.57	2.59	3.66
L3	73.23	73.72	73.92	76.53	1.66	1.39	4.38
L4	68.66	68.58	68.59	71.03	0.85	1.40	3.20
L5	54.64	53.11	53.33	54.50	1.80	1.61	2.23
L6	126.00	120.68	121.19	123.83	6.89	6.25	5.08
L7	165.60	163.59	164.80	163.14	5.70	3.83	4.88
L8	171.65	168.20	170.40	169.49	6.03	2.71	3.11
L9	160.70	158.14	160.53	158.61	5.67	3.53	3.87
L10	137.65	132.88	136.49	136.45	6.11	2.68	4.12
L11	34.42	33.67	36.25	35.24	2.98	3.40	3.00
L12	65.17	64.46	68.18	67.65	4.82	4.19	4.24
L13	56.22	54.25	57.37	60.30	5.14	6.34	5.94
L14	97.49	94.96	97.49	91.75	4.62	1.15	7.35
B1	80.40	75.75	78.45	78.94	5.29	2.32	1.98
B2	58.17	52.08	57.56	57.95	6.85	2.94	2.30

matched with the patterns obtained from DM. Method IM-III is therefore considered as a suitable, accurate and reliable approach of taking hand anthropometric data for the development of custom-made glove patterns.

The findings reveal that even no significant difference exist amongst the four measuring methods on hand circumference and hand length, they tend to produce significant and systematic differences on hand breath and finger dimensions. The dimension differences not only affect the fitting and design of gloves, but also hand functional performance such as finger sensitivity and mobility when gloves are worn. To develop high quality custom-made pressure therapy gloves that perfectly fit the wearer's hands, designers and manufacturers have to understand more the true values of hand anthropometry data by adopting an efficient and accurate measuring method in order to achieve better fitting and performance.

5. Conclusions

This study has revealed important differences in manual hand measurements, and 2D and 3D image analyses. The results indicate that there are significant hand dimension differences amongst the studied four methods. The correlation coefficients indicate that each method is highly linearly correlated with one another which demonstrate that the differences might be systemic and could be reduced by the use of a linear regression model.

Amongst them, Method IM-I which uses a flat colour scanner to extract hand dimensions is the simplest method for practical and medical uses. Nevertheless, it has major differences in comparison to the other three methods. The results may be primarily attributed to the poor contact between the plaster hand model and the scanner bed, leading to major differences in both length and circumference measurements. To pursue high precision hand anthropometric data for making good fitting gloves, it is essential to establish prediction models and/or algorithms for improved accuracy.

The 3D image analysis methods are able to obtain results that are close to direct measurements. The analysis results show that 3D image analysis can provide hand measurements that have no significant differences with manual hand measurements, including hand and finger circumferences, length and breadth dimensions. 3D image analysis has high precision and good repeatability, even though the number of captures has been reduced from 10 captures in Method IM-II to 3 captures in Method IM-III. Method IM-III is therefore considered as the most desirable approach of taking hand anthropometric data since less processing time is required and adverse hand movements during the scanning process could also be minimised.

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